Scenario-9

Problem statement

The client just upgraded their SSL certificates used for the inter broker communication. The cluster was healthy before the certificate updates. After the certificate updates, the client sees the following error in the broker logs -

```
Caused by: java.util.concurrent.CompletionException:
org.apache.kafka.common.errors.TopicAuthorizationException: Not authorized to access topics:
[_confluent-metadata-auth]
        at java.base/java.util.concurrent.CompletableFuture.encodeRelay(CompletableFuture.java:367)
        at java.base/java.util.concurrent.CompletableFuture.completeRelay(CompletableFuture.java:376)
        at java.base/java.util.concurrent.CompletableFuture$AnyOf.tryFire(CompletableFuture.java:1663)
        at java.base/java.util.concurrent.CompletableFuture.postComplete(CompletableFuture.java:506)
        at java.base/java.util.concurrent.CompletableFuture.completeExceptionally(CompletableFuture.java:2088)
        at io.confluent.security.auth.provider.ConfluentProvider.lambda$null$10(ConfluentProvider.java:543)
        at java.base/java.util.concurrent.CompletableFuture.uniExceptionally(CompletableFuture.java:986)
        at
java.base/java.util.concurrent.CompletableFuture$UniExceptionally.tryFire(CompletableFuture.java:970)
        at java.base/java.util.concurrent.CompletableFuture.postComplete(CompletableFuture.java:506)
        at java.base/java.util.concurrent.CompletableFuture.completeExceptionally(CompletableFuture.java:2088)
        at io.confluent.security.store.kafka.clients.KafkaReader.lambda$start$1(KafkaReader.java:102)
        at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:515)
        at java.base/java.util.concurrent.FutureTask.run(FutureTask.java:264)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
```

Observation

There is an issue with the newly created SSL certificate.

Root cause analysis

The issue persists only after creating new SSL certificates, hence lets create new SSL certificates and configure the brokers to use them.

Solution to debug the issue

1. Create a new Certificate Authority (CA) using OpenSSL request

→openssl req -new -x509 -keyout ca-key.pem -out ca-cert.pem -days 365 -subj "/C=IN/ST=Karnataka/L=Uttarahalli/O=Platformatory/OU=Kafka/CN=Platformatory-CA"

2. Creating new certificates for kafka broker 1

→keytool -keystore kafka1.server.keystore.jks -alias kafka1 -validity 365 -genkey -keyalg RSA -dname "CN=kafka1, OU=Kafka, O=Platformatory, L=Uttarahalli, ST=Karnataka, C=IN"

→keytool -keystore kafka1.server.keystore.jks -alias kafka1 -certreq -file kafka1-cert-file

→openssl x509 -req -CA ca-cert.pem -CAkey ca-key.pem -in kafka1-cert-file -out kafka1-cert-signed.pem -days 365 -CAcreateserial

→keytool -keystore kafka1.server.keystore.jks -alias CARoot -import -file ca-cert.pem

→keytool -keystore kafka1.server.keystore.jks -alias kafka1 -import -file kafka1-cert-signed.pem

→openssl verify -CAfile ca-cert.pem kafka1-cert-signed.pem
Verify and expect a reply - #kafka1-cert-signed.pem: OK

→keytool -importcert -file ca-cert.pem -alias CARoot -keystore kafka1.server.truststore.jks

3. Creating new certificates for kafka broker 3

→keytool -keystore kafka2.server.keystore.jks -alias kafka2 -validity 365 -genkey -keyalg RSA -dname "CN=kafka2, OU=Kafka, O=Platformatory, L=Uttarahalli, ST=Karnataka, C=IN"

→keytool -keystore kafka2.server.keystore.jks -alias kafka2 -certreq -file kafka2-cert-file

→openssl x509 -req -CA ca-cert.pem -CAkey ca-key.pem -in kafka2-cert-file -out kafka2-cert-signed.pem -days 365 -CAcreateserial

→keytool -keystore kafka2.server.keystore.jks -alias CARoot -import -file ca-cert.pem

→keytool -keystore kafka2.server.keystore.jks -alias kafka2 -import -file kafka2-cert-signed.pem

→openssl verify -CAfile ca-cert.pem kafka2-cert-signed.pem
Verify and expect a reply - #kafka2-cert-signed.pem: OK

→keytool -importcert -file ca-cert.pem -alias CARoot -keystore kafka2.server.truststore.jks

4. Creating new certificates for kafka broker 3

→keytool -keystore kafka3.server.keystore.jks -alias kafka3 -validity 365 -genkey -keyalg RSA -dname "CN=kafka3, OU=Kafka, O=Platformatory, L=Uttarahalli, ST=Karnataka, C=IN"

→keytool -keystore kafka3.server.keystore.jks -alias kafka3 -certreq -file kafka3-cert-file

→openssl x509 -req -CA ca-cert.pem -CAkey ca-key.pem -in kafka3-cert-file -out kafka3-cert-signed.pem -days 365 -CAcreateserial

→keytool -keystore kafka3.server.keystore.jks -alias CARoot -import -file ca-cert.pem

→keytool -keystore kafka3.server.keystore.jks -alias kafka3 -import -file kafka3-cert-signed.pem

→openssl verify -CAfile ca-cert.pem kafka3-cert-signed.pem
Verify and expect a reply - #kafka3-cert-signed.pem: OK

→keytool -importcert -file ca-cert.pem -alias CARoot -keystore kafka3.server.truststore.jks

Note: Move the CA certificate and its key and also the keystores and truststore files to their respective location on their respective brokers (Change name if necessary).

5. Test the newly generated certificates by running the following command

→ kafka-topics --list --bootstrap-server kafka1:19092 --command-config /opt/client/client.properties

```
nandan@nandan-virtual-machine:~/scenarios/scenario9/cp-sandbox$ docker compose ps -a
WARN[0000] The "ADMIN_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "ADMIN_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] /home/nandan/scenarios/scenario9/cp-sandbox/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME              IMAGE                                          COMMAND                  SERVICE           CREATED          STATUS          PORTS
connect           confluentinc/cp-server-connect:7.4.0           "connect-distributed…"   connect           32 minutes ago   Up 32 minutes   0.0.0.0:8083->8083/tcp, :::8083->8083/tcp, 9092/tcp
control-center    confluentinc/cp-enterprise-control-center:7.4.0   "control-center-star…"   control-center    32 minutes ago   Up 26 minutes   0.0.0.0:9021->9021/tcp, :::9021->9021/tcp
grafana           grafana/grafana:8.5.24                         "/run.sh"                grafana           32 minutes ago   Up 32 minutes   0.0.0.0:3000->3000/tcp, :::3000->3000/tcp
kafka1            confluentinc/cp-server:7.4.0                   "kafka-server-start …"   kafka1            32 minutes ago   Up 32 minutes   9092/tcp
kafka2            confluentinc/cp-server:7.4.0                   "kafka-server-start …"   kafka2            32 minutes ago   Up 32 minutes   9092/tcp
kafka3            confluentinc/cp-server:7.4.0                   "kafka-server-start …"   kafka3            32 minutes ago   Up 32 minutes   9092/tcp
kfkclient         confluentinc/cp-server:7.4.0                   "sleep infinity"         kfkclient         32 minutes ago   Up 32 minutes   9092/tcp
openldap          osixia/openldap:1.3.0                          "/container/tool/run…"   openldap          32 minutes ago   Up 32 minutes   0.0.0.0:389->389/tcp, :::389->389/tcp, 636/tcp
prometheus        prom/prometheus:v2.37.7                        "/bin/prometheus --c…"   prometheus        32 minutes ago   Up 32 minutes   0.0.0.0:9090->9090/tcp, :::9090->9090/tcp
schema-registry   confluentinc/cp-schema-registry:7.4.0          "schema-registry-sta…"   schema-registry   32 minutes ago   Up 28 minutes   0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
zookeeper1        confluentinc/cp-zookeeper:7.4.0                "zookeeper-server-st…"   zookeeper1        32 minutes ago   Up 32 minutes   2888/tcp, 0.0.0.0:2181->2181/tcp, :::2181->2181/tcp, 3888/tc
p
nandan@nandan-virtual-machine:~/scenarios/scenario9/cp-sandbox$ docker compose exec -it -u root kfkclient /bin/bash
WARN[0000] The "ADMIN_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "ADMIN_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] /home/nandan/scenarios/scenario9/cp-sandbox/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[root@kfkclient appuser]# kafka-topics --list --bootstrap-server kafka1:19092 --command-config /opt/client/client.properties
__consumer_offsets
_confluent-command
_confluent-controlcenter-7-4-0-1-AlertHistoryStore-changelog
_confluent-controlcenter-7-4-0-1-AlertHistoryStore-repartition
_confluent-controlcenter-7-4-0-1-Group-ONE_MINUTE-changelog
_confluent-controlcenter-7-4-0-1-Group-ONE_MINUTE-repartition
_confluent-controlcenter-7-4-0-1-Group-THREE_HOURS-changelog
_confluent-controlcenter-7-4-0-1-Group-THREE_HOURS-repartition
_confluent-controlcenter-7-4-0-1-KSTREAM-OUTEROTHER-0000000106-store-changelog
_confluent-controlcenter-7-4-0-1-KSTREAM-OUTEROTHER-0000000106-store-repartition
```

Observation

All the containers are configured to use the newly generated SSL certificates and all the containers are up and running, thus it can list all the topics

Conclusion

Since the previously created certificates had some errors while creating them, we just delete the old certificates and replace them with the newly created ones.